THE FOURTH INTEGRAL BETTI NUMBER OF A TORUS-GRID RIPS COMPLEX

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Abstract. We compute the fourth integral Betti number of the closed radius-five Vietoris–Rips complex on the thirteen-by-thirteen torus grid with its periodic Manhattan metric. It is two, giving a counterexample to the integer-rank formula in AIM Problem 531, which predicts twenty-six for this grid. The proof separates a subcomplex with the homotopy type of the torus from the simplices that cannot be lifted to the plane. A signed integer chain certificate computes the resulting relative homology. Its nonzero even boundary coefficients explain the difference from the published mod-two data.

1. The coefficient distinction

For a periodic grid, the Vietoris–Rips construction can detect topology beyond that of the underlying torus. Adams, Adetowubo, Barriga-Acosta, Feng and Sterling [1] determine several infinite families of torus-grid Rips complexes and tabulate homology in the intermediate-scale range. Their Question 8.4 concerns grids of side length $3k-2$ at scale k. The tables use coefficients in $\mathbb{Z}/2$, whereas their section 3.6 defines Betti numbers as ranks of integral homology. This distinction matters in the integral formulation of AIM Problem 531 [3].

For $n \geq 3$, let $T_n = (\mathbb{Z}/n\mathbb{Z})^2$, with

$$d_n((a,b),(c,e)) = \delta_n(a,c) + \delta_n(b,e), \qquad \delta_n(a,c) = \min\{|a-c|, n-|a-c|\},$$

where coordinates in the last formula lie in $\{0,\dots,n-1\}$. Let $K_n(r)$ consist of all subsets of T_n of diameter at most r. Thus equality with the scale is included. Write $\beta_i(K) = \operatorname{rank} H_i(K;\mathbb{Z})$.

The integral assertion in Problem 531 includes, for every integer $k \geq 5$,

$$\beta_4(K_{3k-2}(k)) = 6k - 3 + (-1)^k. \tag{1}$$

The following single admissible case disproves that universal assertion.

Theorem 1. *For the closed complex $K = K_{13}(5)$ with the metric above,*

$$\operatorname{rank} H_4(K;\mathbb{Z}) = 2.$$

In particular, (1) *fails at $k = 5$, where its right-hand side is* 26.

The proof is computer-assisted. It uses an exact finite certificate, with complete enumeration and integer elementary operations, together with a geometric identification of a torus subcomplex. The certificate and a separate reconstruction and replay program accompany the paper. The result concerns the integral assertion at this one parameter; the mod-two pattern in Question 8.4 remains a separate question.

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2. A subcomplex with the homotopy type of the torus

From now on $n = 13$ and the scale is five. Let $\pi : \mathbb{R}^2 \to \mathbb{R}^2/(13\mathbb{Z})^2$ be the quotient map, and give the plane and torus their Manhattan metrics. Define $M \subseteq K$ to contain those simplices whose vertices have lifts to $\mathbb{Z}^2$ of diameter at most five. Taking subsets preserves this property, so M is a subcomplex.

Lemma 2. *There is a homotopy equivalence $M \simeq \mathbb{R}^2/(13\mathbb{Z})^2$. Consequently,*

$$H_3(M;\mathbb{Z}) = H_4(M;\mathbb{Z}) = 0, \qquad H_4(K;\mathbb{Z}) \cong H_4(K,M;\mathbb{Z}).$$

Proof. Use the open radius-three torus balls centered at the 169 grid vertices. They cover the torus, since every point has a grid vertex at distance at most one. We first identify their nerve with M.

In the plane, set $u = x + y$ and $v = x - y$. For two points, Manhattan distance equals $\max\{|\Delta u|, |\Delta v|\}$. A set of integer points has diameter at most five precisely when both coordinate ranges have width at most five. In that case the intervals

$$(\max u - 3, \min u + 3), \qquad (\max v - 3, \min v + 3)$$

are nonempty, so its open radius-three balls have a common point. Conversely, a common point gives pairwise distance less than six, which is at most five for integer points.

Now take a nonempty intersection of torus balls, choose one grid center and fix its plane lift z_0. Every point of the intersection has a unique lift w with $\|w - z_0\|_1 < 3$. For each other center choose a lift z_i with $\|z_i - w\|_1 < 3$. Then $\|z_i - z_0\|_1 < 6$. This lift of the center is independent of w: two different lifts are at least thirteen apart, while two such choices would be less than twelve apart. Thus the torus intersection is exactly the projection of the fixed convex intersection $\bigcap_i B(z_i, 3)$. Projection is injective there, because it lies in $B(z_0, 3)$. Every nonempty intersection is therefore contractible, and the preceding plane observation identifies the nerve with M. The open-cover nerve theorem [1, Theorem 3.1] gives the claimed homotopy equivalence.

The torus has a two-dimensional CW structure. Homotopy invariance, cellular homology and the comparison with simplicial homology give the two vanishing statements [2, Corollary 2.11 and Theorems 2.27, 2.35]. For completeness, the fourth-homology isomorphism can be seen directly on quotient chains. A relative four-cycle lifts to a four-chain whose boundary is a three-cycle in M; subtracting a four-chain in M with that boundary gives an absolute cycle. If an absolute cycle is a relative boundary, subtract that boundary to leave a four-cycle in M, which bounds in M. These operations prove surjectivity and injectivity. □

We also need an explicit way to recognize M in a finite enumeration. It is generated by the 338 projected closed diamonds of radius $5/2$ with centers

$$(a + \tfrac{1}{2}, b), \qquad (a, b + \tfrac{1}{2}), \qquad a, b \in \mathbb{Z}/13\mathbb{Z}. \tag{2}$$

Indeed, for a lifted simplex choose the center in (u, v) coordinates to be $(\min u + 5/2, \min v + 5/2)$. Both coordinate ranges lie within distance $5/2$ of this center. The corresponding (x, y) center has one integral and one half-integral coordinate, so it is of the form (2). Conversely, every such diamond has diameter five. This is the scale-five instance of the planar facet geometry in [1, Proposition 4.2]; the argument above proves the membership test used here.

3. The signed relative chain calculation

We use the relative simplicial chain groups $C_q(K, M;\mathbb{Z}) = C_q(K;\mathbb{Z})/C_q(M;\mathbb{Z})$. Their basis consists of the simplices of K that are not in M. Label a vertex (a, b) by $13a + b$, orient every simplex by increasing labels, and order the basis lexicographically in each degree. Its boundary is the alternating sum of faces, omitting those in M.

3.1. **Complete enumeration.** Every edge of K has a unique shortest plane displacement, because its length is at most five and $13 > 10$. Fix a vertex of a clique as anchor and lift its other vertices by these displacements. The clique lies in M if and only if all pairwise distances of these lifts are at most five. If the test fails, a triangle containing the anchor has nonzero total edge displacement in $(13\mathbb{Z})^2$. To see this, suppose every anchored triangle has zero displacement sum. Then every difference between two anchored lifts is the unique displacement of their edge and has length at most five, contrary to failure of the test.

This gives a finite exhaustive enumeration: list all nonliftable triangles through $(0, 0)$, extend each by all cliques in its common-neighbor set, translate by every element of T_{13}, and remove duplicates. Every nonliftable simplex contains one of these translated triangles, and every extension remains nonliftable. There is no degree cutoff in this enumeration. It returns 24 origin triangles, at most eight possible extension vertices per triangle, and no nonliftable simplex with more than seven vertices.

An alternative reconstruction starts with the full neighbor graph and recursively lists cliques through $(0, 0)$ in increasing label order. It tests membership in the diamonds (2), without using the triangle criterion. A branch is skipped only when the union of the current clique and all its possible extension vertices lies in one diamond. Every descendant then belongs to M. Translating the retained cliques gives exactly the same relative bases.

Finite verification procedure. The input is the closed metric graph on T_{13} and the fixed cancellation certificate. The output is its relative fourth-homology free rank.

(1) Enumerate all noncore cliques by either reconstruction above.
(2) Form the signed relative boundary matrices in the prescribed bases, and check that every consecutive boundary composition is zero.
(3) Replay the recorded unit cancellations, checking each pivot and both adjacent chain identities.
(4) Compare the complete residual integer matrix and count its nonzero columns; compute its rank modulo three.
(5) Use the matching minor and column bounds to determine its rational rank, and subtract that rank from the residual fourth chain rank.

The triangle construction considers at most $24 \cdot 2^8$ anchored extension subsets and 169 translates per subset before deduplication. The stored relative basis has 12168 simplices, each with at most seven vertices, and each initial boundary has at most seven terms. The recorded elimination has 6056 unit pivots. These are bounds and counts for the fixed input, not a complexity assertion for arbitrary torus grids.

3.2. **Integral cancellation.** The following elementary fact accounts for every elimination step.

Lemma 3. *In a chain complex of finite free abelian groups, a boundary entry equal to* 1 *or* -1 *can be cancelled. If, after permuting bases, that boundary is*

$$A = \begin{pmatrix} a & \beta \\ \gamma & D \end{pmatrix}, \qquad a \in \{1, -1\},$$

the remaining boundary is $D - \gamma a^{-1}\beta$. *The resulting chain complex has the same integral homology.*

Proof. The integer invertible matrices

$$L = \begin{pmatrix} a^{-1} & 0 \\ -\gamma a^{-1} & I \end{pmatrix}, \qquad S = \begin{pmatrix} 1 & -a^{-1}\beta \\ 0 & I \end{pmatrix}$$

give $LAS = \operatorname{diag}(1, D - \gamma a^{-1}\beta)$. Change the adjacent differentials by the corresponding inverse basis changes. The chain identities force the coefficient in the distinguished upper-boundary row and the distinguished lower-boundary column to vanish. The changed complex therefore splits off a two-term summand $\mathbb{Z} \xrightarrow{1} \mathbb{Z}$, whose homology is zero. Deleting this summand proves the assertion. □

In a recorded step, let r be the pivot row and c the pivot column, with entry $a = \pm 1$. A column with entry b in row r is replaced by itself minus (b/a) times column c. Delete r and c, together with their corresponding column and row in the adjacent differentials. Indices remain as labels for the changed bases. The replay program checks that the adjacent directions vanish after the basis changes, as in the lemma. In particular, a coefficient of absolute value two is never used as a unit pivot.

Proposition 4. *The fixed certificate reduces $C_*(K, M; \mathbb{Z})$ to a complex with nonzero chain groups only in degrees $2, 3, 4$, of ranks $2, 28, 26$, respectively. Its only nonzero differential is an integer matrix $A_4 : \mathbb{Z}^{26} \to \mathbb{Z}^{28}$ with exactly 24 nonzero columns and*

$$\operatorname{rank}_{\mathbb{Q}} A_4 = 24.$$

Proof. Complete reconstruction gives the initial chain ranks in Table 1. Signed boundary squares vanish over $\mathbb{Z}$. The two enumerations agree on every basis simplex, and the independent replay checks all 6056 pivots of the certificate using Lemma 3. The counts in the table leave exactly the stated residual chain groups.

TABLE 1. Exact relative chain ranks and recorded unit cancellations. A degree-q cancellation removes one generator in degrees q and $q - 1$.

Degree q	Initial rank	Degree-q cancellations	Residual rank
2	1352	0	2
3	4394	1350	28
4	4394	3016	26
5	1690	1352	0
6	338	338	0

The certificate contains the entire residual 28×26 matrix. All its nonzero entries are 2 or -2, and exactly 24 columns are nonzero. This bounds its rational rank above by 24. Exact row reduction modulo three gives rank 24; consequently some 24×24 integral minor is nonzero modulo three, hence nonzero over $\mathbb{Q}$. This gives the matching lower bound and proves the proposition. □

Proof of Theorem 1. After the homology-preserving cancellations, the fifth chain group is zero, so $H_4(K, M; \mathbb{Z})$ is the integral kernel of A_4. The rational span of this kernel equals $\ker(A_4 : \mathbb{Q}^{26} \to \mathbb{Q}^{28})$: clear denominators in any rational kernel vector. Thus its free rank is $26 - 24 = 2$. Lemma 2 transfers this rank to $H_4(K; \mathbb{Z})$. Since $13 = 3 \cdot 5 - 2$ and $6 \cdot 5 - 3 + (-1)^5 = 26$, this is an admissible counterexample to (1). □

REPRODUCIBILITY

The accompanying files contain a deterministic generator, `src/torusgrid_certificate.py`, the independent reconstruction and replay program `src/verify_certificate.py`, and the proof-critical input `numerical_case/certificates/k5_certificate.json`. Python 3.10 or later with its standard library suffices. From the artifact root, run

```
python3 src/verify_certificate.py \
  numerical_case/certificates/k5_certificate.json
python3 -m unittest discover -s test -v
python3 numerical_case/generate_certificate.py \
  --output /tmp/k5-regenerated.json
```

The generator requires a fresh output path. Its output contains no timestamps or platform-dependent values and can be compared directly with the fixed certificate. The verifier imports no generator code. It checks the finite calculation; the geometric core and the transfer to the original complex are proved above. The tests include a four-sphere boundary and a triangulation of the real projective plane, checking in particular that even boundary coefficients are preserved. The zero reduction of A_4 modulo two accounts for the larger fourth dimension in the mod-two calculation.

Automated assistance

OpenAI Codex was used for proof exploration, exact verification software, source checking and manuscript preparation. The argument and certificate are submitted for independent mathematical review.

References

[1] H. Adams, A. Y. Adetowubo, H. Barriga-Acosta, Z. Feng and J. Sterling, *Vietoris–Rips Complexes of Torus Grids*, Mediterranean Journal of Mathematics **22** (2025), article 173, doi:10.1007/s00009-025-02945-9. Author version: arXiv:2502.07134v2.

[2] A. Hatcher, *Algebraic Topology*, Cambridge University Press, 2002.

[3] *AIM problem collection*, GitHub repository `MColbrook/AIM`, Problem 531, "Homology at the second transition of torus-grid Rips complexes", revision `a153f469ee0ea64e876b2ad97c79d252c19a3e22`, immutable problem statement.

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